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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Security Attacks, Threats, Risks and Their Impact on Security Goals

CSA5116-CRYPTOGRAPHY AND NETWORK SECURITY

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INTRODUCTION TO INFORMATION SECURITY

- **What are Security Goals?**

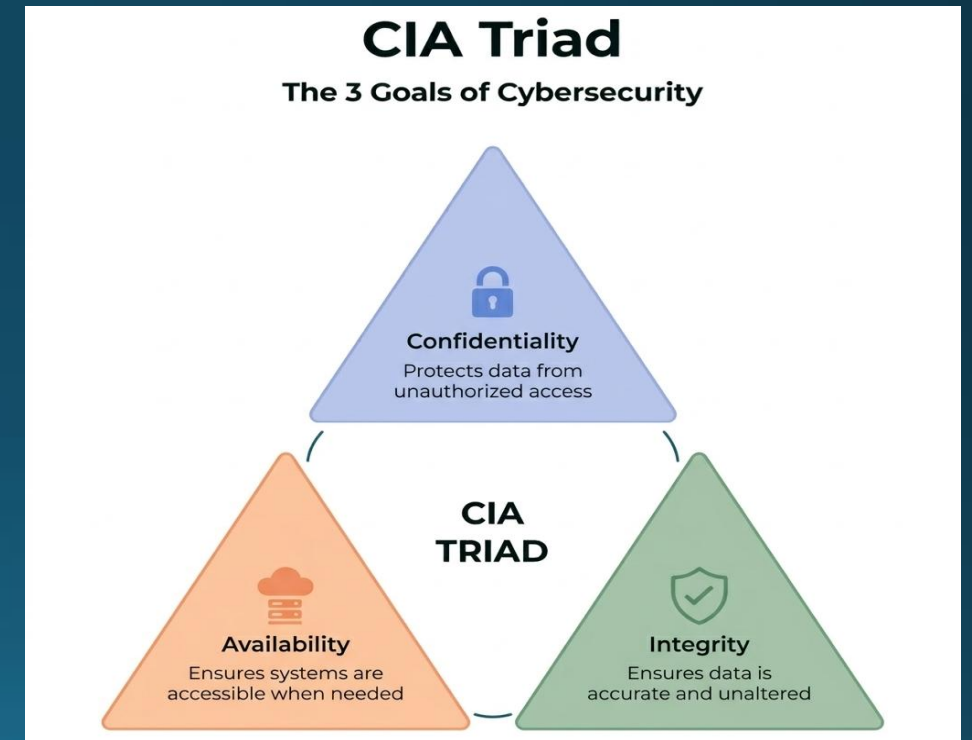
Security goals are the fundamental principles that protect information and ensure secure communication.

- **Why are They Important?**

- Protect sensitive information
- Prevent unauthorized access
- Ensure reliable and secure systems
- Build trust and ensure compliance

Core Security Goals

- **Confidentiality** – Protects data from unauthorized access.
- **Integrity** – Ensures data remains accurate and unaltered.
- **Availability** – Ensures authorized users can access information when needed.



UNDERSTANDING THREATS, ATTACKS & RISKS

➤ Threat

- A potential danger that can harm a system or its information.
- **Examples:** Hacker, Malware, Natural Disaster

➤ Attack

- An intentional attempt to exploit a system and gain unauthorized access.
- **Examples:** Phishing, Ransomware, DoS Attack

➤ Risk

- The chance of loss or damage when a threat exploits a vulnerability.
- **Examples:** Data theft, Financial loss, Service disruption

Threat → Vulnerability → Attack → Risk → Impact

- Hacker (Threat) → Phishing Email (Attack) → Stolen Password (Risk)

TYPES OF SECURITY ATTACKS

Passive Attacks

- *Observe information without altering it.*
- **Release of Message Contents** – Unauthorized reading of data.
- **Traffic Analysis** – Monitoring communication patterns.

Active Attacks

- *Modify, disrupt, or destroy information.*
- **Masquerade** – Impersonating an authorized user.
- **Replay Attack** – Resending intercepted data.
- **Message Modification** – Altering transmitted messages.
- **Denial of Service (DoS)** – Disrupting access to services.

Passive Attack	Active Attack
▪ Observes data	▪ Modifies or disrupts data
▪ Difficult to detect	▪ Easier to detect
▪ Affects Confidentiality	▪ Affects Integrity & Availability

SECURITY MEASURES

➤ Encryption

Protects sensitive information by converting it into unreadable cipher text.

➤ Authentication

Verifies the identity of users before allowing access to systems or data.

➤ Access Control

Restricts access to information based on user roles and permissions.

➤ Firewalls

Monitors and filters network traffic to prevent unauthorized access.

➤ Multi-Factor Authentication (MFA)

Strengthens security by requiring two or more verification methods.

➤ Backup & Recovery

Ensures important data can be restored after cyberattacks or system failures.



ENCRYPTION

Converts data into unreadable format to protect it from unauthorized access.



AUTHENTICATION

Verifies the identity of users before granting access to systems or data.



ACCESS CONTROL

Restricts access to resources based on user roles and permissions.



FIREWALLS

Monitors and filters network traffic to block unauthorized access and threats.



MULTI-FACTOR AUTHENTICATION (MFA)

Adds an extra layer of security by requiring two or more verification methods.



DATA BACKUP & RECOVERY

Creates copies of important data and restores it in case of loss, damage, or cyberattacks.



IMPACT OF SECURITY THREATS

Data Breaches

- Unauthorized access to confidential or sensitive information.
- **Financial Loss**
- Costs incurred due to cyberattacks, fraud, ransom payments, and system recovery.

Operational Disruption

- Interruptions in business operations caused by attacks such as malware or Denial-of-Service (DoS).

Identity Theft

- Personal or organizational information is stolen and misused for fraudulent activities.

Reputational Damage

- Loss of customer trust, reduced credibility, and negative impact on an organization's image.



THANKYOU